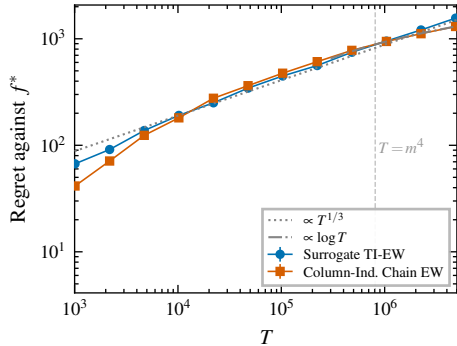
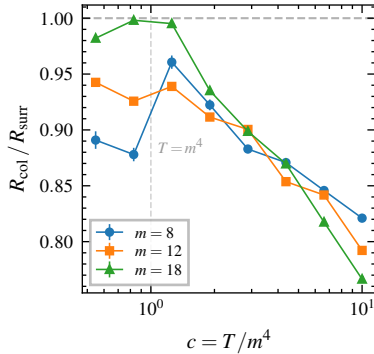


(a) $m = 30$ 

(b) Regret ratio

(c) Horizon-free, $m = 12$ 